1) problem statement

- A university wants to automate its examination process

a) pseudocode

```
START
 CLASS person
    name, age
 CLASS student EXTENDS person
  regNo, marks[5]
 - Average()
 CLASS ResearchStudent EXTENDS
  Student
      researchArea
      DisplayDetails()
 Main
      CREATE ResearchStudent R
      INPUT details
      R.Average()
      R.DisplayDetails()
END
```

(2) Complete the blanks to remove duplicate elements from an integer array

```
BEGIN
READ N
READ ARRAY A
FOR i ← 0 TO N-1
    FOR j ← i+1 TO N-1
        IF A[i] == A[j]
            REMOVE A[j]
        ENDIF
    END FOR
ENDFOR
PRINT Modified Array
END
```

3 The following pseudocode should calculate the area of both a square and a rectangle using overloaded methods.

Ⓐ <u>pseudocode:</u>

```
CLASS Area
  method area(Length)
     RETURN Length * Length
 END METHOD
 METHOD area (Length, breadth)
    RETURN Length * breadth
 END METHOD
 END CLASS
```

④ predict the output

Ⓐ

- Start : Sum = 0
- i = 1 (odd) → sum = 0 - 1 = -1
- i = 2 (even) → sum = -1 + 2 = 1
- i = 3 (odd) → sum = 1 - 3 = -2
- i = 4 (even) → sum = -2 + 4 = 2
- i = 5 (odd) → sum = 2 - 5 = -3

*output:

-3

5

problem statement

A

```
BEGIN
READ N
READ A [0 ..., N-1]
first ← -∞
second ← -∞
for i ← 0 TO N-1
    if A[i] > first Then
        Second ← first
        first ← A[i]
    ELSE if A[i] > second and A[i] ≠ first then
        second ← A[i]
    end if
endFOR
print second
end
```